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L_RAG: A Controlled Ablation Study of Vietnamese Legal Question Answering

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Final Project Presentation

Outline

Why Vietnamese Legal QA Is Difficult

Legal requirements

- Answers must be grounded in authoritative statutes.
- Citations must identify the correct article, clause, and item.
- Current validity and amendment history must be respected.
- Many questions require evidence across documents.

Why standard RAG is insufficient

- Dense search may miss exact statutory terminology.
- Fixed chunks break document hierarchy.
- Semantic similarity ignores legal precedence.
- LLMs may hallucinate or cite expired provisions.

Research problem

How can semantic matching, exact lexical search, statutory structure, temporal validity, and reranking be combined in one reliable legal RAG pipeline?

L_RAG: Objectives

1. Build an end-to-end hybrid pipeline combining dense E5 retrieval, BM25, and Reciprocal Rank Fusion.
2. Construct an in-memory legal knowledge graph for hierarchy, reading order, validity, authority, and context expansion.
3. Integrate multilingual rerankers: mMiniLMv2, BGE-Reranker-v2-m3, Jina-Reranker-v2, and Qwen3-Reranker-0.6B.
4. Generate answers from selected evidence using `gpt-4o-mini`, with strict citation and abstention rules.
5. Construct a standardized benchmark of 500 Vietnamese legal QA pairs.

Target output

A natural-language answer supported by traceable legal chunks, provision-level citations, authority metadata, and validity context.

Main Contributions

Complete pipeline

Modular ingestion, structuring, indexing, hybrid retrieval, graph traversal, reranking, and generation.

Dynamic legal graph

Query-time validity overlays and precedence sorting without rebuilding the vector index.

Benchmarking stack

A 500-query benchmark and support for four multilingual reranking backends with GPU-safe sequential loading.

Core contribution: evidence that is semantically relevant, structurally complete, temporally valid, and traceable.

Retrieval-Augmented Generation

Standard RAG

1. Encode the user query.
2. Retrieve semantically similar chunks.
3. Add retrieved evidence to the LLM prompt.
4. Generate a grounded answer.

Domain limitation

- Semantic ambiguity lowers precision.
- Fixed-size chunks lose long-range structure.
- Self-reflection methods do not directly model legal authority or validity.

Standard RAG reduces unsupported generation, but legal QA needs explicit statutory structure.

Legal Information Retrieval

Properties of legal text

- Deep hierarchy: code, chapter, article, clause, item.
- High sensitivity to exceptions and exact wording.
- Formal precedence among legal instruments.
- Dynamic amendments, replacements, and repeals.

Implication for retrieval

- A topically similar chunk may be legally inapplicable.
- Local evidence may require parent or sibling context.
- Validity depends on both time and document relationships.
- Neural embeddings alone cannot encode every explicit dependency.

Design requirement

The retriever must combine text relevance with authority, validity, hierarchy, and provenance.

Hybrid Retrieval, Reranking, and GraphRAG

Dense + sparse

E5 captures paraphrases; BM25 captures exact article numbers, legal codes, and domain terms.

Rank fusion

RRF combines heterogeneous ranked lists without requiring directly comparable scores:

$$\mathrm{RRF}(d) = \sum_{m} \frac{1}{60 + r_m(d)}$$

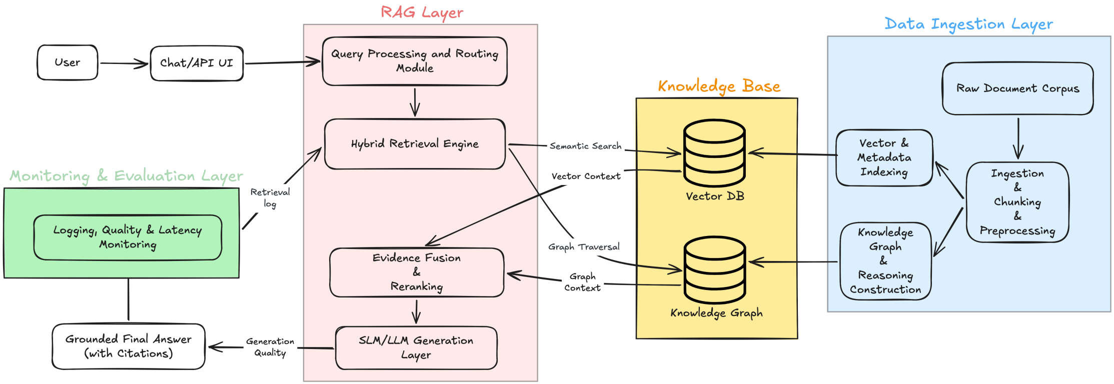
Graph + reranking

Graph traversal restores related legal context; cross-encoders jointly score each query–chunk pair.

L_RAG synthesis

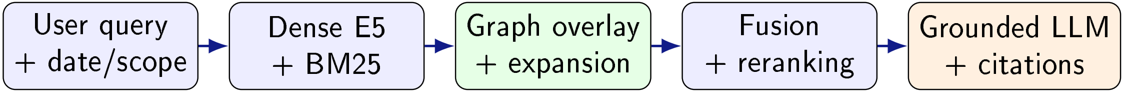
Use dense and sparse retrieval for coverage, graph operations for legal coherence, and cross-encoders for final precision.

End-to-End L_RAG Architecture



Offline: corpus structuring, vector indexing, and graph construction. Online: hybrid retrieval, graph overlay, reranking, and cited generation.

Five-Stage Online Pipeline



1. Retrieve a broad candidate pool from complementary dense and sparse indexes.
2. Apply legal whitelists and expand connected structural context.
3. Fuse and rerank candidates, then send the top 10 chunks to the generator.

Dense and Sparse Retrieval

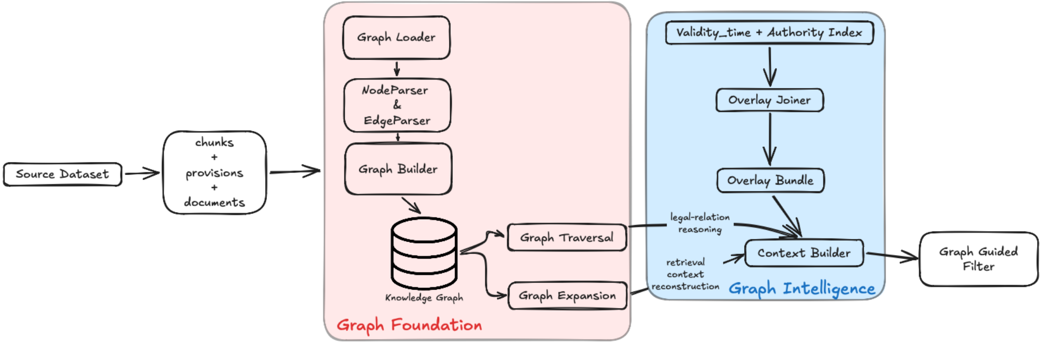
Dense vector search

- Model: `multilingual-e5-large` (560M, 1024 dimensions).
- FAISS `IndexFlatIP` over L2-normalized vectors.
- Embedding input combines document title, provision header, and chunk content.
- Strength: semantic paraphrases and natural-language questions.

Sparse keyword search

- BM25 with $k_1 = 1.5$ and $b = 0.75$.
- Vietnamese tokenization using `underthesea`.
- Strength: exact legal terms, document numbers, and citations.
- Dense and sparse lists are fused using RRF ($k = 60$).

Knowledge Graph Overlay



The graph coordinates document filtering, validity reasoning, precedence resolution, and reading-order context expansion.

Knowledge Graph Operations

Query-time reasoning

- Fold validity events relative to `as_of_date`.
- Resolve conflicts by legal authority and version.
- Produce `id_str_filter` whitelists for retrieval.
- Traverse only direction-verified legal edges.

Structural expansion

- `DOCUMENT_HAS_PROVISION`
- `PROVISION_HAS_CHUNK`
- `CHUNK_NEXT`
- `PROVISION_NEXT`

Sibling chunks are expanded in reading order to restore missing local context.

Implementation status

The in-memory loader, builder, traversal, expansion, overlay, and context modules pass all 27 knowledge-graph tests.

Sequential Multi-Reranker Stage

Supported backends

1. mMiniLMv2 (117M): lightweight baseline.
2. BGE-Reranker-v2-m3 (567M): multilingual reranker.
3. Jina-Reranker-v2 (278M): multilingual long-context model.
4. Qwen3-Reranker-0.6B: causal-LM reranker in FP16.

GPU-safe execution

$$\mathrm{Load}(M_i) \to \mathrm{Run}(M_i)$$

$$\to \mathrm{Unload}(M_i) \to \mathrm{empty\_cache}()$$

Candidates: $K = 30$
Final evidence: $N = 10$

Why rerank?

Unlike separate bi-encoder vectors, a cross-encoder jointly attends to the complete query and candidate passage.

Citation-Grounded Answer Generation

Generator configuration

- Model: `gpt-4o-mini`
- Temperature: 0.0
- Maximum output: 1,024 tokens
- Context: top 10 reranked chunks

Prompt contract

- Cite governing legal provisions explicitly.
- Use only retrieved evidence.
- Return `INSUFFICIENT_CONTEXT` when evidence is inadequate.
- Format boolean, extractive, abstractive, and unanswerable outputs appropriately.

The LLM synthesizes evidence; it is not treated as the legal source.

Vietnamese Legal Corpus at a Glance

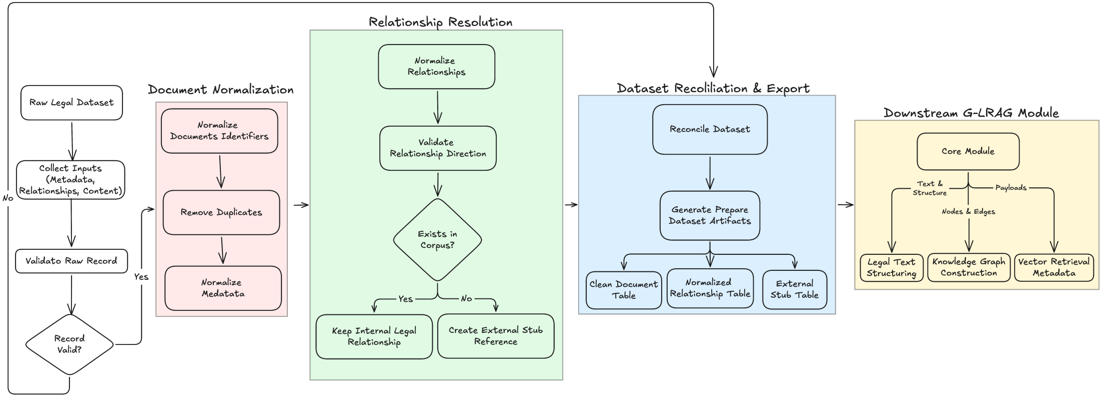
Entity	Count
Legal documents	151,624
Hierarchical provisions	1,386,267
Retrieval chunks	1,513,376
Document relationships	883,256
External referenced stubs	19,763

Corpus coverage

The collection includes National Assembly laws, Government decrees, Prime Minister decisions, and ministry circulars.

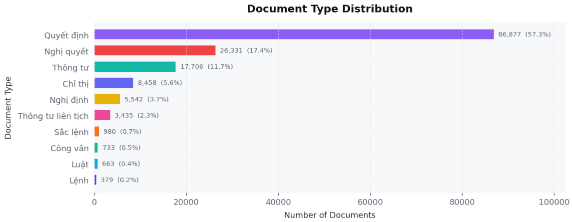
Each chunk retains document identity, provision identity, title, issuing authority, effective date, and provenance for retrieval and citation.

Corpus Preparation Pipeline

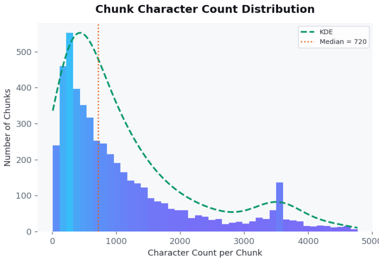


Raw metadata, relationships, and HTML content are normalized, structured into legal units, enriched with overlays, and exported for vector and graph indexing.

Document and Chunk Characteristics



Document-type distribution



Chunk-length distribution

Chunk design

The processed corpus contains 1.51M structural retrieval chunks. Chunk text is enriched with document title and provision header before embedding.

Knowledge Graph Schema

Node types

- **Document**: authority, issuer, effective date, validity.
- **Provision**: article/clause header and unit type.
- **Chunk**: retrieval text and sequence number.
- **External stub**: referenced entity outside the corpus.

Edge types

- Containment: document $\rightarrow$ provision $\rightarrow$ chunk.
- Reading order: `CHUNK_NEXT`, `PROVISION_NEXT`.
- Legal dependencies: `AMENDS`, `REPLACES`, `CITES`.
- Only verified directions may be traversed.

Integrity gating

Quarantined records and unverified edge directions are excluded from trusted traversal to reduce graph poisoning.

500-Query Legal QA Benchmark

Answerability

- 400 answerable questions
- 100 unanswerable questions
- Unanswerable cases test hallucination resistance and abstention.

Answer types

- Abstractive
- Boolean
- Extractive
- Unanswerable

Question categories

- Citation
- Cross-document
- Legal validity
- Multi-hop
- Single-hop

Difficulty levels: easy, medium, and hard.

Ground Truth and Evaluation Contract

For each answerable question, the benchmark stores:

$$Q_i = \{\mathrm{qa\_id},\ \mathrm{question},\ \mathrm{category},\ \mathrm{difficulty},\ \mathrm{answer\_type},\ \mathrm{ref\_answer},\ \mathrm{gold\_chunks}\}$$

Retrieval metrics

- Recall@k and Hit@k
- MRR@k and nDCG@k
- Precision@k / chunk overlap

Generation metrics

- Exact Match
- Token F1 and ROUGE-L
- Unanswerable accuracy
- Citation grounding and faithfulness

Gold chunk IDs connect each benchmark answer directly to retrievable legal evidence.

Controlled Experimental Protocol

Shared settings

- Frozen benchmark: 500 QA pairs.
- Retrieval metrics: 400 answerable cases.
- Dense encoder: `multilingual-e5-large`.
- Generator: `gpt-4o-mini`, $T = 0$, 1,024 tokens.
- Comparisons paired by `qa_id`.

Uncertainty analysis

- Mean per-question deltas.
- Percentile-bootstrap 95% confidence intervals.
- 10,000 resamples for primary comparisons.
- A CI containing zero is treated as inconclusive.

Important

Candidate budgets differ across experiment families, so scores are compared **within a family**, not as one pooled factorial study.

Three Completed Ablation Families

Family	Compared configurations	Candidate/final	Changed factor
Embedding text	Chunk-only; chunk + metadata	10/10	Indexed text
Retrieval/graph	DenseOnly; HybridOnly; DenseGraph; HybridGraph	50/10	BM25 + graph
Reranking	None; RRF; mMiniLM; Qwen3; BGE; Jina	30/10	Final ranker

Fixed candidate-pool test

All rerankers receive the same reconstructed RRF pool of 30 chunks. Candidate Recall@30 is 0.5995 and Hit@30 is 0.6450 for every configuration.

The planned generator-reasoning ablation is not included because semantic legal-correctness labels are not yet available.

Ablation 1: Does Metadata Improve Embeddings?

Variant	Hit@10	Recall@10	MRR@10	Document@10
Chunk-only	53.5%	48.5%	28.6%	65.8%
Chunk + metadata	53.2%	48.5%	22.7%	70.2%
Paired delta	−0.2 pp	+0.1 pp	−5.9 pp	+4.5 pp

Positive effect

Document success improves by **+4.5 pp**; 95% CI $[+0.5, +8.8]$.

Trade-off

Exact-chunk MRR falls by **5.9 pp**; 95% CI $[-9.1, -2.7]$.

Metadata helps locate the correct document, but the first exact gold chunk appears later.

Ablation 2: Dense, Hybrid, and Graph Retrieval

Method	Recall@10	MRR@10	nDCG@10	ID precision	Avg. chunks	Total p50 (s)
DenseOnly	**0.4855**	**0.2267**	**0.2813**	0.0580	10.000	5.483
HybridOnly	0.4257	0.2252	0.2638	0.0493	10.000	31.538
DenseGraph	0.3615	0.2045	0.2374	**0.0800**	5.605	**4.998**
HybridGraph	0.3076	0.2012	0.2228	0.0641	6.305	31.180

Hybrid effect

- Recall changes by −5.98 pp vs. DenseOnly.
- 95% CI: [−9.71, −2.21].
- Median latency rises by about 26 seconds.

Graph effect

- Recall: −12.40 pp (dense), −11.81 pp (hybrid).
- Higher identity precision, but lower evidence coverage.
- Expansion removes more gold evidence than it adds.

Ablation 3: Reranker Comparison

Model	Recall@10	Hit@10	MRR@10	nDCG@10	Token F1	ROUGE-L	Rerank p50	Total p50
None	0.4842	0.5375	0.2167	0.2714	0.2801	0.2453	0.000	2.925
RRF-only	0.4382	0.4800	0.2335	0.2728	0.2750	0.2416	0.000	2.972
mMiniLM	0.4973	0.5425	0.3047	0.3425	0.2901	0.2568	**0.385**	3.343
Qwen3	0.5218	0.5725	0.3418	0.3688	0.2912	0.2583	9.695	13.007
BGE	0.5188	0.5675	0.3247	0.3593	**0.2955**	**0.2603**	2.872	5.848
Jina	**0.5305**	**0.5850**	**0.3670**	**0.3914**	0.2913	0.2558	1.278	4.158

Best retrieval frontier

Jina gives the best Recall, Hit, MRR, and nDCG with intermediate reranking cost.

Other operating points

BGE slightly leads lexical answer overlap; mMiniLM is the fastest learned reranker.

Why Jina Is the Final Reranker Candidate

Paired gain vs. RRF-only

- Recall@10: **+9.23 pp**
  CI $[+5.51, +12.98]$
- MRR@10: **+13.35 pp**
  CI $[+9.51, +17.24]$
- nDCG@10: **+11.85 pp**
  CI $[+8.46, +15.27]$

Largest category gains

- Legal validity ($n = 91$): Recall **+14.47 pp**
- Citation ($n = 106$): Recall **+11.89 pp**
- Abstractive ($n = 199$): Recall **+13.65 pp**
- Boolean ($n = 123$): Recall **+7.86 pp**

Evidence ceiling

Jina retains 88.50% of gold chunks available in the shared candidate pool, compared with 71.29% for RRF-only. The gain is better promotion, not a larger pool.

Efficiency and What the Metrics Do Not Prove

Measured efficiency

- Dense total p50: 5.483 s.
- Hybrid total p50: 31.538 s.
- Reranker p50: mMiniLM 0.385 s; Jina 1.278 s; BGE 2.872 s; Qwen3 9.695 s.
- Cached BM25 makes reranker latency incomplete for full-online deployment.

Missing validation

- Human-rated legal correctness and completeness.
- Claim faithfulness to retrieved passages.
- Citation precision and citation recall.
- Full-online p95 latency, throughput, and GPU memory.

Interpretation boundary

Token F1, ROUGE-L, and refusal markers are diagnostics; they cannot certify that an answer is legally correct.

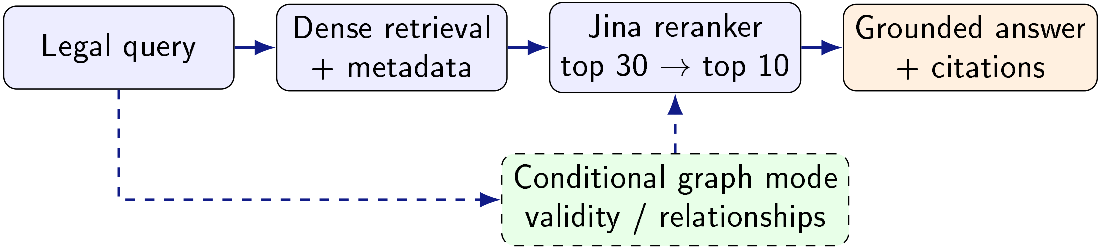
Interpretation and Design Implications

1. **Metadata changes the level of success:** it improves document identification, but can push the exact gold chunk lower in the ranking.
2. **Current hybrid retrieval is not a good default:** its paired recall loss and large latency cost outweigh the observed benefit under the tested settings.
3. **Current graph expansion trades coverage for density:** adjacent passages can replace high-confidence seed evidence within a fixed context budget.
4. **Reranking is the clearest positive result:** every tested cross-encoder improves Recall, MRR, and nDCG over RRF-only at a fixed candidate pool.

What can still change?

These findings apply to the tested index, tokenization, fusion constant, candidate budgets, graph policy, and benchmark—not to every hybrid or GraphRAG design.

Recommended Engineering Direction



Default path

- Dense retrieval followed by Jina.
- mMiniLM when latency is the priority.
- BGE when lexical overlap is preferred.

Conditional modules

- Retune BM25/RRF before enabling hybrid by default.
- Gate graph expansion by query intent.
- Preserve high-confidence seed passages.

Limitations and Ethical Considerations

Threats to validity

- Only 500 questions; multi-hop, hard, and cross-document slices are small.
- Gold chunks may omit legally sufficient alternatives.
- Candidate budgets differ across experiment families.
- One encoder, generator, prompt, and corpus version were tested.
- Graph relationships still contain direction and validity uncertainty.

Responsible use

- Expose citations and validity metadata.
- Abstain when evidence is insufficient.
- Protect privacy in user questions.
- Account for uneven legal-corpus coverage.
- Require qualified human legal review.

Conclusion and Future Work

Final conclusion

The controlled study supports **dense retrieval followed by a cross-encoder**, with **Jina** as the current quality–latency candidate. Metadata helps at document level; the tested hybrid and graph policies require redesign before becoming defaults.

Next experiments

– Collect blinded labels for legal correctness, completeness, faithfulness, and citation quality.

– Sweep candidate-pool size, RRF parameters, and graph-context budgets.

– Implement seed-preserving graph expansion and validate edge directions.

– Evaluate generator reasoning strategies and larger, more diverse benchmarks.

– Measure full-online p95 latency, GPU memory, and throughput.

The results are benchmark-bounded engineering evidence—not certification of legal reliability.

Sources

- `docs/report/ablation_report.tex`: updated final ACL-style report entry point.
- `docs/report/sections/`: system, experiments, results, discussion, limitations, and conclusion.
- Paired per-question analyses and run manifests referenced by the final report.
- L_RAG system, dataset, graph, retrieval, and benchmark specifications.

Questions & Discussion

Thank you!